

ARNAV NEGI

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Skills

- **Programming:** Python, SQL (MySQL, Oracle PL/SQL), Bash, C, C++, MATLAB
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, PyTorch, Model Evaluation, Ensemble Methods, Deep Learning
- **Computer Vision:** Semantic Segmentation, Segment Anything (SAM), SegFormer, DeepLabV3+, OpenCV
- **Data Analysis:** Pandas, NumPy, EDA, Data Cleaning, Feature Engineering, Geospatial Analysis
- **NLP & LLMs:** Transformers, Hugging Face, LangChain, LangGraph, Information Retrieval, RAG Systems
- **Cloud & Systems:** AWS, Docker, Cloud Computing
- **Web Development:** Django, Bootstrap
- **Visualization:** Matplotlib, Streamlit, Folium
- **Tools:** Git, GitHub, Jupyter, Google Colab, VS Code
- **Soft Skills:** Leadership, Stakeholder Engagement, Public Speaking, Cross-cultural Communication

Education

University of Queensland

Jul 2024 – Jul 2026

Master of Computer Science (Management)

Brisbane, Australia

- Relevant Coursework: Machine Learning, Artificial Intelligence (Search & Planning), Natural Language Processing, Information Retrieval, Data Mining, Cloud Computing

Thapar Institute of Engineering and Technology

Aug 2020 – Jun 2024

Bachelor of Engineering in Computer Engineering

Punjab, India

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks

Professional Experience

APS (Association of Postgraduate Students)

Dec 2025 – Jul 2026

President (Coursework) - Part-time Leadership Role

Brisbane, Australia

- Represented a cohort of 10,000+ postgraduate students by serving in a paid leadership role across university governance and committee forums.
- Delivered 8–10 student engagement events per semester reaching 500+ attendees by leading a team of 15+ student representatives.
- Collaborated with 10+ academic, administrative, and student stakeholders to deliver initiatives across academic support, wellbeing, and career development.
- Improved participation in student programs by ~25% through targeted outreach initiatives and redesigned engagement strategies.

University of Queensland (UQ)

Jan 2026 – Feb 2026

Data Science Intern – Genomic Prediction & Machine Learning

Brisbane, Australia

- Improved genomic prediction accuracy across datasets containing 10,000+ SNP features by designing and evaluating XGBoost, Random Forest, and Neural Network models.
- Improved model evaluation efficiency by ~30% by engineering scalable data pipelines and automating large-scale genomic analyses.
- Identified optimal modelling strategies by benchmarking machine learning, deep learning, and GBLUP approaches using RMSE and Pearson correlation metrics.
- Enhanced predictive robustness by applying ensemble learning techniques across multiple modelling approaches.

Integra Micro Systems

Jun 2023 – Aug 2023

Software Development Intern

Bengaluru, India

- Improved information retrieval efficiency by ~40% across 10,000+ documents by developing LangChain-powered chatbot systems and hybrid search pipelines.
- Improved response relevance for domain-specific queries by integrating keyword and semantic retrieval techniques into a hybrid search architecture.
- Reduced response latency by ~25% by optimizing prompt engineering and response generation workflows.

Technical Projects

CoralReef AI - Multimodal RAG Chatbot for Coral Bleaching Education

- Improved accessibility of coral bleaching education by building a multimodal AI chatbot combining scientific retrieval and image-supported responses.
- Increased answer transparency and traceability by implementing a LangGraph workflow incorporating retrieval, reasoning, generation, and trace logging.
- Created a domain-specific environmental knowledge base by processing and indexing 13,000+ knowledge chunks from research PDFs, web articles, and conservation resources.
- Reduced unsupported AI responses by implementing a retrieval-augmented generation pipeline with vector search and metadata-aware retrieval.
- Improved user understanding of reef health indicators by integrating coral imagery with evidence-grounded textual explanations.

Dwell-Aware TravelPlanner - Temporal Feasibility Modelling for AI Itinerary Generation

- Improved temporal realism of LLM-generated itineraries by extending the OSU-NLP TravelPlanner benchmark with dwell-time-aware planning.
- Generated 3,300+ POI-level dwell-time labels by building a large-scale estimation pipeline across 1.3M+ user check-ins and 143k+ points of interest.
- Improved attraction duration modelling by aggregating 117k+ clean activity episodes across tourism, shopping, and food categories.
- Enhanced itinerary context awareness by integrating Google Places, OpenStreetMap, and routing-derived accessibility and travel-time features.
- Evaluated temporal feasibility across 386 itinerary days by designing custom metrics including overloaded-day rate, balanced-day rate, and plan-level feasibility.
- Quantified the impact of dwell-aware planning by reproducing and benchmarking against the GPT-3.5 TravelPlanner baseline.

ResearchDoc - AI-Powered Research Management Platform

- Streamlined research project management by developing a Django/MySQL platform supporting project workflows, resource management, and AI-assisted summarization.
- Reduced manual synthesis effort by implementing an OpenAI-powered pipeline enabling users to generate, review, save, and manage structured research summaries.
- Improved platform scalability and maintainability by designing modular Django architectures supporting project-resource relationships and subscription archiving.

StreetView Urban Features - Semantic Segmentation for Urban Spatial Analysis

- Built a street-view computer vision pipeline using Mapillary imagery and Cityscapes-pretrained semantic segmentation models to quantify urban spatial characteristics.
- Extracted urban indicators including Building Coverage Ratio, Sky Openness Ratio, Green View Index, Pedestrian Space Ratio, and Vehicle Space Ratio from pixel-level segmentation masks.
- Benchmarked semantic segmentation performance on the Cityscapes validation set and applied the pipeline to spatially sampled street-view imagery for local urban analysis.

SAM Prompt Sensitivity - Prompt-Conditioned Image Segmentation Analysis

- Designed controlled experiments to evaluate how Meta's Segment Anything Model (SAM ViT-B) responds to point, bounding-box, and automatic mask prompting strategies.
- Quantified prompt sensitivity using pairwise mask IoU, observing mean agreement of 0.637 for point vs. box prompts, 0.483 for point vs. automatic masks, and 0.463 for box vs. automatic masks.
- Built a reproducible PyTorch and OpenCV evaluation pipeline for GPU-based segmentation, mask comparison, visualization, and prompt-sensitivity analysis.

Leadership, Activities & Engagement

UQ Business School Postgraduate Association

Feb 2025 – Dec 2025

Director of Corporate Relationships (Executive)

Brisbane, Australia

- Organized 4+ networking and corporate engagement events with 100–150 attendees each, connecting students with industry professionals.
- Coordinated with 5+ sponsors and external partners to support event execution, marketing, and outreach initiatives.
- Strengthened postgraduate participation in professional development programs through targeted communication and stakeholder engagement.

APS (Association of Postgraduate Students)

Jul 2025 – Dec 2025

International Officer

Brisbane, Australia

- Led cultural engagement initiatives supporting a diverse international student community of 5,000+ students.
- Developed and managed an events calendar covering 6+ major cultural festivals, increasing event participation by ~30%.
- Coordinated with cross-functional student teams (10+ members) to deliver inclusive events fostering representation and cross-cultural dialogue.

University of Queensland

Nov 2025

Leaders at EAIT Pathway Program

Brisbane, Australia

- Completed a competitive leadership development program involving 20+ hours of training in self-leadership, mentoring, and diversity.
- Applied leadership frameworks in real-world student engagement contexts through workshops and collaborative initiatives.

Additional Achievements & Activities

University of Queensland

Jul 2024-Jul 2025

Campus Kitchen Volunteer

Brisbane, Australia

- Supported meal preparation and service for 100+ students per session in a fast-paced, team-oriented environment.
- Collaborated with diverse teams, strengthening cross-cultural communication and teamwork skills.

UQ Union

Jul 2024-Jul 2025

Cloth Shop Volunteer Assistant

Brisbane, Australia

- Assisted in retail operations including inventory management, customer interaction, and store organization.
- Developed professional communication and customer service skills through engagement with a diverse student community.

Pinegrove School

2013 – 2018

Student Leader & Activities

Himachal Pradesh, India

- Completed a 12-day high-altitude cycling expedition from Manali to Ladakh, summiting Khardung La (18,380 ft) and Tanglang La (17,580 ft), among the world's highest motorable passes.
- Summited Himalayan peaks including Chandarnahan (~13,980 ft), Batseri Kanda (~19,000 ft), and Deo Tibba (6,001 m / ~19,688 ft).
- Winner of Inter-State Padmashree Declamation Contest, Shimla, demonstrating strong public speaking skills.
- Selected to present the school report before the Chief Minister of Himachal Pradesh.
- Awarded school "Colors" for excellence in adventure sports and co-curricular achievements.